

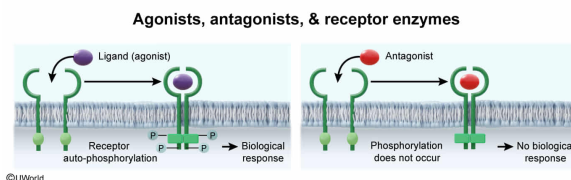
Sorafenib is an antagonist that binds the extracellular domain of colony stimulating factor 1 receptor (csfr), a receptor tyrosine kinase that, when overly activated, can cause encephalomyelitis (inflammation of the brain and spinal cord). In what way could sorafenib best help relieve encephalomyelitis?

- ☐ A. It could inhibit formation of an action potential across neuronal cell membranes. [10%]
- ✓ ☒ B. It could prevent the intracellular domains of csfr proteins from phosphorylating each other. [82%]
- ☐ C. It could increase dimerization of csfr proteins. [4%]
- ☐ D. It could reduce degradation of csfr proteins by cytosolic proteases. [1%]

Correct

83% answered correctly

Explanation:



Receptor enzymes are **transmembrane proteins** that catalyze chemical reactions in response to extracellular signals. Upon ligand binding to the extracellular domain of the protein, a **conformational change** occurs that allows the intracellular domain to become catalytically active. **Antagonists** are molecules that bind to receptor proteins and thereby **reduce the overall response to the normal ligand** (ie, agonist), either through competition for the same binding site or by other means.

Receptor tyrosine kinases (RTKs) are one of the most common types of receptor enzymes. Upon ligand binding, RTKs **dimerize**. Each RTK in the dimer then phosphorylates one or more tyrosine residues on the intracellular domain of the other RTK. This action ultimately activates a pathway that alters the cell's behavior.

The question states that csfr is an RTK that can cause encephalomyelitis when overly active. Sorafenib is an antagonist that binds csfr. As such, sorafenib binding would likely **inhibit csfr proteins from phosphorylating other csfr intracellular domains**. This would reduce the activity of csfr, which may relieve encephalomyelitis.

(Choice A) **Action potentials** are rapid voltage changes across a membrane that are conducted along the membrane and result from the activities of **ion channels** and **pumps**. Csfr is not an ion channel or a pump and is most likely not directly involved in formation of action potentials.

(Choice C) Binding of an antagonist would most likely either *decrease* dimerization of

csfr proteins or have no effect on dimerization levels. Antagonist binding is unlikely to increase dimerization.

(Choice D) Antagonists do not typically alter degradation rates of the proteins to which they bind; they simply inhibit the normal biological response. In addition, if sorafenib *did* reduce degradation, more csfr proteins would be available for activation, so reduced csfr degradation would most likely *aggravate* encephalomyelitis rather than relieve it.

Educational objective:

Receptor enzymes are transmembrane proteins that become catalytically active upon ligand binding. Receptor tyrosine kinases are common receptor enzymes that dimerize and phosphorylate each other when ligands bind. Antagonists bind to receptor enzymes without inducing a biological response.

Time spent:0 sec

Subject:
Biochemistry

QID:403408

System:
Enzymes